F697 — Task #6 (lives-there): honest bucket-classification of the mixing VALUES for the partition table. The mixing MECHANISM is bucket-1 containment for all — every mixing value IS a two-point Born overlap of two FORCED stratifications (flag = Korányi–Wolf boundary orbits, frame = spectral idempotents), hence a functional of μ (genus). The SPECIES question (is the *exact value* a forced functional?) splits: $\sin^2\theta_{13}$ = 1/45 and |sin δ_PMNS| = 2/7 are species-FORCED (full bucket-1) — the Pythagorean LAW g^2 = 45 + 4 = N_c^2·n_C + rank^2 forces both numerator and denominator (already the flagship's 3 LAW rows). $\sin^2\theta_{12}$ = 3/10 and $\sin^2\theta_{23}$ = 4/7 stay species-IDENTIFIED (bucket-1 genus, species open): the DENOMINATORS are homed as rep-dimensions (10 = dim SO(5), 7 = dim ℝ^{5,2}) but the NUMERATORS {N_c, rank^2} are UNPINNED (K763, ~8 routes). There is a candidate numerator-route for θ_{12} — $\sin^2\theta_{12}$ = dim(V_{12})/dim(SO(5)) = N_c/10, with V_{12} the Peirce off-diagonal (dim a = N_c) that connects the two Cartan idempotents in the Jordan decomposition V = ℝc_1 ⊕ V_{12} ⊕ ℝc_2 — but this is exactly the "intertwiner = the connecting space" step K763 flagged as *asserted, not proven*, so I do NOT bank it and do NOT back-solve. δ_CKM = arctan($\sqrt{\text{n\_C}}$) is VALUE-SPEC (a rank-2 exact identity; the CP phase is the phase of the overlaps — genus bucket-1, species not forced). Net for #6: the mechanism (mixing = Born overlap of forced strata) is bucket-1 containment for all four; θ_{13} + δ_PMNS-magnitude move fully into bucket-1 species; θ_{12} + θ_{23} stay species-identified (numerator forcings candidate-but-unproven, K763); δ_CKM value-spec. Honest partition status — what is forced moves, what is not stays labeled, no back-solving.

Lyra, Sat 2026-07-25 13:19 EDT. #6 lives-there: the disciplined bucket-classification of the mixing values. Moves what's forced (2 of 4 species) fully to bucket-1; the mechanism is bucket-1 for all; holds the 2 unpinned numerators honestly (candidate route flagged, K763-unproven, not back-solved).

The classification (for the partition table #7)

mixing value	closed form	genus (containment)	species (forced?)	note
$\sin^2\theta_{13}$	1/(N_c^2·n_C) = 1/(g^2−rank^2) = 1/45	Born overlap (B↔S) ⍰	FORCED — bucket-1	Pythagorean LAW g^2=45+4; num=1, den=45 both forced
|sin δ_PMNS|	rank/g = 2/7	phase of overlap ⍰	FORCED — bucket-1	same LAW ($\sin^2$+$\cos^2$=1 exact); sign observed

mixing value	closed form	genus (containment)	species (forced?)	note
$\sin^2\theta_{23}$	rank^2/g = 4/7	Born overlap (I↔S) ⍰	IDENTIFIED	den 7 = dim ℝ^{5,2} homed; num rank^2=4 unpinned (K763)
$\sin^2\theta_{12}$	N_c/(rank·n_C) = 3/10	Born overlap (B↔I) ⍰	IDENTIFIED	den 10 = dim SO(5) homed; num N_c=3 candidate (Peirce V_{12}) but K763-unproven
δ_CKM	arctan($\sqrt{}$n_C)	phase of overlap ⍰	VALUE-SPEC	rank-2 exact identity; J_CKM~3e-5; species not forced

The candidate numerator-route for θ_{12} (surfaced, NOT banked)

The three angles are the three pairs of the 3 KW strata (bulk B / Cartan-slice I / Shilov S): θ_{12} = B↔I, θ_{23} = I↔S, θ_{13} = B↔S. A structural reading of θ_{12}'s numerator: - The B↔I (bulk↔Cartan) transition crosses the Peirce off-diagonal V_{12} (the space connecting the two Cartan idempotents in V = $\mathbb{R}c_1 \oplus V_{12} \oplus \mathbb{R}c_2$), which has dim a = N_c (the short-root/color space, K865/T2511). - So $\sin^2\theta_{12}$ = dim(V_{12})/dim(SO(5)) = N_c/10 — the fraction of the SO(5) structure that is the connecting off-diagonal. This is a candidate for homing the numerator N_c — but it IS the "intertwiner = the connecting space" step K763 flagged as asserted-not-proven. So I present it as a lead, not a bank: if the two-point overlap genuinely forces the connecting-dimension = dim(V_{12}), θ_{12} moves to bucket-1 species; until then it stays identified. No back-solving.

Honest #6 verdict

- Bucket-1 (genus/containment) — ALL four: mixing = Born overlap of the forced flag and frame stratifications; every value is a functional of μ. Solid, exhibited.
- Bucket-1 (species/forced) — 2 of 4: $\sin^2\theta_{13}$ = 1/45, |sin δ_PMNS| = 2/7 (the Pythagorean LAW). Fully derived.
- Species-identified — 2 of 4: $\sin^2\theta_{12}$ = 3/10, $\sin^2\theta_{23}$ = 4/7 — denominators homed (rep-dims), numerators candidate-but-unproven (K763). δ_CKM value-spec.
- The open piece (what would move θ_{12}/θ_{23} to bucket-1 species): the two-point overlap forcing the numerator as a connecting-dimension (V_{12} = N_c for θ_{12}; the analog for θ_{23}'s rank^2). This is a real, sourceable computation — but it was pursued ~8 routes (K763) without closing, so I flag it as the standing open species, not a fresh win.

Tiers / handoffs

- @Keeper (#6, #7, #8) — lives-there classified for the partition table: mixing MECHANISM = bucket-1 containment for all four (Born overlap of forced flag/frame strata); θ_{13} + δ_PMNS-mag species-FORCED (LAW, full bucket-1); θ_{12} + θ_{23} species-IDENTIFIED (denominators homed as rep-dims, numerators candidate-but-unproven per K763); δ_CKM value-spec. I surfaced but did NOT bank a candidate θ_{12} numerator-route (Peirce V_{12} dim = N_c) — it's the K763 "asserted-not-proven" step. For exhaustiveness: the mixing rows are bucket-1-genus (containment holds), with 2/4 species forced. Log honestly; don't count θ_{12}/θ_{23} as species-forced.
- @Grace (#2, #7) — for the containment table + SM table: the mixing values are all Born-overlap functionals of μ (containment), and I've classified the species status (2 forced, 2 identified, δ_CKM value-spec). The candidate θ_{12} route (dim V_{12}/dim SO(5) = N_c/10) is worth a check against your K763 record — is it one of the ~8 that failed, or a route not yet closed? That decides whether θ_{12} can move to species-forced.

- @Casey — worked the "lives-there" question for the mixing angles — do their exact values *derive* as forced functionals, or only *match*? Honest answer: the *mechanism* is fully derived and bucket-1 — every mixing angle is a genuine geometric overlap between two ways the domain stratifies itself, so it IS a functional of the one measure. And two of the four exact values are forced outright (the reactor angle θ_{13} and the PMNS phase magnitude, both riding the same Pythagorean identity $g^2 = 45 + 4$). The other two (the solar angle 3/10 and the atmospheric 4/7) have their *denominators* pinned as honest dimensions of the domain's structures, but their *numerators* aren't forced yet — there's a tempting reading where the solar angle's "3" is the color/short-root dimension, but that's exactly the step we tried eight ways last week and couldn't nail, so I'm holding it as a lead, not banking it. So the mixing sector lands like everything else in this table: the structure derived, the mechanism bucket-1, and the exact numbers honestly split into forced and still-identified — no back-solving to inflate the count.

Notes only; no toys/theorems claimed. #6 lives-there: mixing MECHANISM = bucket-1 containment for all (Born overlap of forced flag/frame strata). SPECIES: θ_{13}=1/45 + |sin δ_PMNS|=2/7 FORCED (Pythagorean LAW, full bucket-1); θ_{12}=3/10 + θ_{23}=4/7 IDENTIFIED (denominators homed as rep-dims 10=dim SO(5), 7=dim $\mathbb{R}$^{5,2}; numerators N_c/rank^2 candidate-but-unproven, K763 — θ_{12} candidate route = dim V_{12}/dim SO(5), NOT banked, not back-solved); δ_CKM=arctan($\sqrt{}$n_C) value-spec. 2 of 4 species move to bucket-1; mechanism bucket-1 for all; open piece = two-point overlap forcing the numerators (K763 standing). — Lyra